NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

Requests shall be referred to Headquarters, U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, MD 21005-5055, usarmy.APG.atec.mbx.atec-standards@mail.mil

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13. ABSTRACT

Describes a method for evaluation of heating equipment operational and performance characteristics. Identifies supporting tests, facilities, and equipment required. Provides procedures for heating capacity and smoke tests. Applicable to space, radiant, portable nonduct and duct type heaters.

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14. KEY WORDS	LINK A		LINK B		LINK C	
	ROLE	WT	ROLE	WT	ROLE	WT
Heater						
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Environmental control set						

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REFERENCE FILE

U.S. ARMY TEST AND EVALUATION COMMAND
ENGINEERING TEST - SYSTEM TEST OPERATIONS PROCEDURES

AMSTE-RP-702-109
*Test Operations Procedure 10-2-072 20 April 1972
AD742517

HEATING EQUIPMENT

		Paragraph	Page
Section I.	GENERAL		
	Purpose and Scope	1	1
	Background	2	1
	Equipment and Facilities	3	2
II.	TEST PROCEDURES		
	Supporting Tests	4	2
III.	SUPPLEMENTARY INSTRUCTIONS		
	Capacity	5	4
	Smoke Test	6	4
APPENDIX.	REFERENCES		7

SECTION I
GENERAL

1. Purpose and Scope. This TOP describes test procedures for evaluating the operational and performance characteristics of heating equipment. Equipment covered includes: space, radiant, portable nonduct and duct type heaters. From the tests listed in Section II, the test director can select those that will satisfy the requirements for the particular test item and the particular test type (i.e., engineering test, initial production test, etc.). Test objectives are to determine conformance of the test items to QMR, MN or other suitability criteria. For initial production tests, scope will be in accordance with the contractual provisions of the applicable military specifications and suitability criteria established by the test directive. This document provides for simulated environmental testing but does not include service testing or environmental testing at climatic test sites.

2. Background. The requirement of nonduct heaters to support field operations of the Armed Services entail a variety of problems and a continuing challenge in research. To fulfill these heater requirements, new types of heaters are constantly being developed and the existing heaters are being improved or modified to support the military requirements brought about by the new tactical and logistical concepts of

*This TOP supersedes MTPs 10-2-070 (24 Jul 67), and 10-2-071 (21 Mar 68), including all changes.

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warfare. These new and improved heaters then are the results of projects to correct deficiencies, to improve upon operation, and to reduce cost without changing general operational characteristics. The heaters must be capable of efficiently performing their required function in the field for which they have been developed or modified.

3. Equipment and Facilities. Equipment and facilities required are defined in the documents listed in Section II and Section III.

SECTION II
TEST PROCEDURES

4. Supporting Tests. Subtests (generally in preferred order of completion with respect to high-risk, short duration) to be considered in formulating a test plan are listed below with references.

	TEST SUBJECT TITLE	PUBLICATION NO.
a.	Pre-operational Inspection	10-3-500
	(1) Operator Training and Familiarization	10-2-501
	(2) Photographic Coverage	7-3-519
b.	Physical Characteristics	10-2-500
c.	Safety	10-2-508
d.	Performance	
	(1) Coal Fired Space Heater	MIL-H-11219B Para 4.2
	(a) Capacity (Refer to para 5)	
	(b) Smoke Test (Refer to para 6)	
	(2) Electric Space Heater	MIL-H-19910 Paras 4.5.2, 4.5.3
	Capacity (Refer to para 5)	
	(3) Multifuel Space Heater	MIL-H-52204B Paras 4.5.2.4, 4.5.2.6, 4.5.2.8
	(a) Capacity and Airflow	Para 4.5.2.1
	(b) Controls and Safety Operation	Para 4.5.2.3
	(4) Duct Heaters	MIL-H-52230A Paras 4.5.2.6, 4.5.2.8, 4.5.2.10
	(a) Capacity and Airflow	Para 4.5.2.1
	(b) Controls and Safety Operation	Para 4.5.2.3

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TEST SUBJECT TITLE	PUBLICATION NO.
e. Environmental Testing	
(1) Altitude	MIL-STD-810B Method 500
(2) Temperature	Method 501 AR70-38
(3) Sunshine	4-2-826
(4) Rain	2-2-815
(5) Humidity	4-2-820
(6) Fungus	4-2-818
(7) Salt Fog	MIL-STD-810B Method 509
(8) Dust Test	Method 510
(9) Vibration	4-2-804 MIL-H-52230A Para 4.5.2.2.7
(10) Rough Handling Shock	4-2-602 MIL-H-52204B Para 4.5.2.2.7
(11) Electromagnetic Interference Characteristics	MIL-STD-461A Notice 4 MIL-STD-462 Notice 3 Methods CE02, RE05
f. Transportability	
(1) Road, Rail, Marine	10-2-503
(2) Air	7-2-515
g. Human Factors Evaluation Sound-Level	10-2-505
(1) Space Heater	MIL-H-52204B Para 4.5.2.5
(2) Duct Heater	MIL-H-52230A Para 4.5.2.5
h. Reliability Confidence Intervals and Sampling Size	AMCP 702-3 3-1-002
i. Durability	10-2-502
j. Maintenance Evaluation	10-2-507
k. Value Analysis	USAMC SUPPL 1 to AR 11-26

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SECTION III
SUPPLEMENTARY INSTRUCTIONS

5. Capacity.

a. Objective. To determine the heating characteristics of the test item.

b. Method. The test item is mounted or installed in the center of the test area. Thermocouples or mercury thermometers are positioned about the test item as shown in figure 1. The test item is then operated for one hour at test conditions to insure temperature stabilization and uniform operation. After stabilization, the required data is recorded every 15 minutes for a total of not less than five complete sets of data. These procedures are repeated for test item operating conditions of one-quarter, one-half, three-quarters and maximum output.

c. Data Required.

(1) Test item nomenclature and type.

(2) Temperature recorded at the installed thermocouples or thermometers.

(3) Fuel weight in pounds and ounces or power input in volts, watts and amps used during recorded test periods.

(4) Flue-gas temperature, °F (read within one inch of where flue-gas analysis sample is taken), if applicable.

(5) Flue-gas analysis, if applicable (refer to paragraph 4.5.2.1.6, MIL-H-52204B (appendix)).

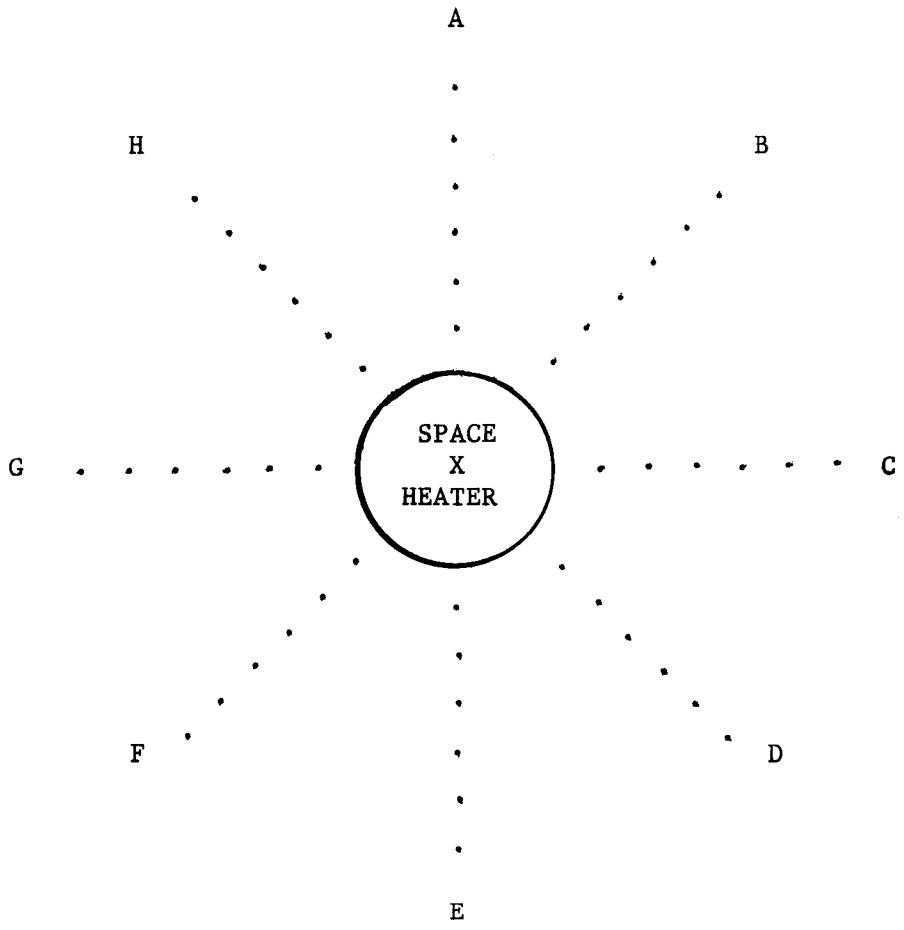
d. Analytical Plan. The indicated and measured outputs and efficiencies are computed and compared with the requirements of the MN to determine conformance to specifications.

6. Smoke Test.

a. Objective. To determine that the test item does not emit smoke which is hazardous to health and welfare of personnel.

b. Method. A sample of gases emitted from the test item is collected and subjected to a laboratory analysis to determine its content. Photographs are taken of any visible gas emissions (smoke). During this test the heater is installed in a tent or other shelter for which it is intended with the normal limitations on ventilation.

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NOTE: 1. Each dot represents an increment/measurement at which a thermocouple or thermometer will be installed away from the heater.

2. "X" is the thermocouple in the flue, when applicable.

3. Drawing is not to scale.

Figure 1. Example of Arrangement of Thermocouples or Thermometers.

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c. Data Required.

(1) Test item nomenclature and type.

(2) Volume of gas sample.

(3) Photographs of visible gas emissions (smoke).

(4) Laboratory analysis of gas sample.

d. Analytical Plan. The contents of the analyzed gas sample and the photography of the visible gas emissions are compared with the requirements of the MN to determine conformance to specifications.

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APPENDIX
REFERENCES

1. AR 70-38, "Research, Development Test, and Evaluation of Materiels for Extreme Climatic Conditions."

2. USAMC Supplement 1 to AR 11-26, "Value Engineering."

3. AMCP 702-3, "Quality Assurance - Reliability Handbook."

4. MIL-STD-461A, "Electromagnetic Interference Characteristics, Requirements for Equipment", including notices 1 thru 4.

5. MIL-STD-462, "Electromagnetic Interference Characteristics, Measurement of", including notices 1 thru 3.

6. MIL-STD-810B, "Environmental Test Methods", including notices 1 thru 4.

7. MIL-H-11219B, "Heater, Space, Non-Electric (Magazine Type, Coal-Fired, 50,000 BTU Per Hour)."

8. MIL-H-19910, "Heater, Space, Electric."

9. MIL-H-52204B, "Heaters, Space; Multi Fuel, With Blower, 15,000 BTU/Hr."

10. MIL-H-52230A, "Heaters, Duct-Type, Portable: 150,000 BTU/Hr."

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